Aquifer, reservoir, plant

This page covers four output families:

- Aquifer storage and fluxes (`aquifer_*.txt`, `basin_aqu_*.txt`).
- Reservoir storage and fluxes (`reservoir_*.txt`, `basin_res_*.txt`).
- Crop yields (`crop_yld_aa.txt`, `crop_yld_yr.txt`, `basin_crop_yld_*.txt`).
- Plant and weather (`hru_pw_*.txt`, `lsunit_pw_*.txt`, `basin_pw_*.txt`).

Aquifer

Source

- Writer: src/aquifer_output.f90, units 2520 (day), 2521 (mon), 2522 (yr), 2523 (aa).
- Basin sum: src/basin_aquifer_output.f90, units 2090-2093.
- File opening: src/header_aquifer.f90, src/header_write.f90.
- Record type: src/aquifer_module.f90, `type aquifer_dynamic`.

Format

Seven leading columns (jday, mon, day, yr, unit, gis_id, name), then the 17 fields of `aquifer_dynamic` in declaration order.

#	Column	Units	Description
1	flo	mm	lateral flow from aquifer
2	dep_wt	m	average depth from surface elevation to water table
3	stor	mm	average water storage in aquifer in timestep
4	rchrg	mm	recharge entering aquifer from other objects
5	seep	mm	seepage from bottom of aquifer
6	revap	mm	plant water uptake plus evaporation from aquifer
7	no3_st	kg/ha N	current total NO3-N mass in aquifer
8	minp	kg/ha P	mineral P transported in return (lateral) flow
9	cbn (header label: `orgn`)	percent	organic carbon in aquifer (static). Header mislabels this column.
10	orgn (header label: `orgp`)	kg/ha N	organic nitrogen in aquifer (static). Header mislabels this column.
11	no3_rchg	kg/ha N	NO3-N flowing into aquifer from another object
12	no3_loss	kg/ha	NO3-N loss
13	no3_lat	kg/ha N	nitrate load to reach in groundwater
14	no3_seep	kg/ha N	nitrate seepage to next object
15	flo_cha	mm	aquifer return flow into channels
16	flo_res	mm	aquifer return flow into reservoirs
17	flo_ls	mm	aquifer return flow into a landscape element

Example header

`refdata/Ames_sub1/basin_aqu_aa.txt`:

```
 demo                      SWAT+ 2024-08-27       MODULAR Rev 2024.61.0.1-33-g68b1328
   jday    mon    day     yr    unit  gis_id  name                          flo          dep_wt             stor
                                                                             mm               m               mm
```

The `aqu_header` strings in `aquifer_module.f90` declare columns `orgn` (position 9) and `orgp` (position 10), but the `aquifer_dynamic` type fields in those positions are `cbn` (organic carbon percent) and `orgn` (organic nitrogen). The writer pushes whole records in declaration order, so the column under the header `orgn` is carbon and the column under `orgp` is organic nitrogen.

Aggregation

- Daily, monthly, yearly, annual average. Aggregation rules: monthly `stor`, `dep_wt`, `no3_st` are divided by the number of days in the month so they remain averages; yearly divides those three by 12; annual average divides everything by `time%yrs_prt`.

Reservoir

Source

- Writer: src/reservoir_output.f90, units 2540 (day), 2541 (mon), 2542 (yr), 2543 (aa).
- Basin sum: src/basin_reservoir_output.f90.
- File opening: src/header_reservoir.f90.
- Record types: `type water_body` in src/water_body_module.f90, `type hyd_output` in src/hydrograph_module.f90.

Format

The writer call is:

```
write (2540,100) time%day, time%mo, time%day_mo, time%yrc, j, ob(iob)%gis_id, ob(iob)%name, &
  res_wat_d(j), res(j), res_in_d(j), res_out_d(j)
```

Seven leading columns, then four blocks. Each `hyd_output` block contributes its 18 fields. `res(j)` is the full `reservoir` derived type and contributes the storage fields written into the `_stor` columns.

Block	Source	Columns
1	`res_wat_d(j)` (`water_body`)	area (ha), precip (m^3), evap (m^3), seep (m^3)
2	`res(j)` storage as `hyd_output` (`_stor` columns)	18 storage fields: flo, sed, orgn, sedp, no3, solp, chla, nh3, no2, cbod, dox, san, sil, cla, sag, lag, grv, temp
3	`res_in_d(j)` (`hyd_output`, `_in` columns)	18 inflow fields
4	`res_out_d(j)` (`hyd_output`, `_out` columns)	18 outflow fields

See Channel and sediment outputs for the 18 fields of `hyd_output` and their units; the layout is identical to the sd_channel writer (inflow is `m^3/s`, outflow is `m^3/s`, storage is `m^3`).

Example header

`refdata/Ames_sub1/basin_res_aa.txt`:

```
 demo                        SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328
   jday    mon    day     yr      unit    gis_id   name                      area          precip            evap
```

The basin-level reservoir output uses `m^3` for the inflow / outflow columns instead of the `m^3/s` used by `basin_sd_cha_aa.txt`, because the reservoir code does not divide by seconds. The header reports `m^3` accordingly.

Crop yield

Source

- Writer: bottom block of src/hru_output.f90, units 4008 (aa) and 4010 (yr). Triggered only at `time%end_sim == 1` (for `_aa`) or `time%end_yr == 1` (for `_yr`).
- Basin variant: `basin_crop_yld_*.txt`, written in the basin output path.
- File opening: src/output_landscape_init.f90.

Format

Different shape from the other output files: each line is one HRU per plant, not one HRU per timestep.

#	Column	Units	Description
1	jday	int	last simulated day
2	mon	int	last simulated month
3	day	int	day of month
4	yr	int	year
5	unit	int	HRU number
6	PLANTNM	char	plant name (`pldb(idp)%plantnm`)
7	MASS	kg/ha	total yield mass (dry weight)
8	C	kg/ha	carbon in yield
9	N	kg/ha	nitrogen in yield
10	P	kg/ha	phosphorus in yield

`crop_yld_aa.txt` reports the per-harvest average (yield divided by number of harvests). `crop_yld_yr.txt` reports the yield for that year.

Example

`refdata/Ames_sub1/crop_yld_aa.txt`:

```
 demo
  SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328

                                                                --YIELD (kg/ha)--
  jday   mon   day    yr    unit PLANTNM                  MASS          C           N           P
   366    12    31  2020       1    corn                                    3358.149    1511.167
   366    12    31  2020       1    soyb                                       0.000       0.000
```

Switched on in `print.prt` by the `crop_yld` row (`a` for annual average only, `y` for yearly only, `b` for both).

Plant and weather

Source

- HRU writer: src/hru_output.f90, units 2040 (day), 2041 (mon), 2042 (yr), 2043 (aa).
- File opening: src/output_landscape_init.f90.
- Record type: src/output_landscape_module.f90, `type output_plantweather`.

Format

Seven leading columns, then the 25 fields of `output_plantweather` in declaration order.

#	Column	Units	Description
1	lai	$m^{2/m}2$	leaf area index
2	bioms	kg/ha	total plant biomass
3	yield	kg/ha	harvested biomass yield (dry weight)
4	residue	kg/ha	surface residue cover
5	sol_tmp	deg C	temperature of soil layer 2
6	strsw	days	water stress (drought)
7	strsa	days	excess water stress (aeration)
8	strstmp	days	temperature stress
9	strsn	days	nitrogen stress
10	strsp	days	phosphorus stress
11	strss	days	salinity stress
12	nplnt	kg N/ha	plant N uptake
13	percn	kg N/ha	NO3-N leached past soil profile

#	Column	Units	Description
14	pplnt	kg P/ha	plant P uptake
15	tmx	deg C	maximum air temperature
16	tmn	deg C	minimum air temperature
17	tmpav	deg C	average air temperature
18	solrad	MJ/m^2	solar radiation
19	wndspd	m/s	windspeed
20	rhum	none	relative humidity
21	phubase0	deg C/deg C	base-zero potential heat units
22	lai_max	$m^{2/m}2$	maximum LAI during timestep
23	bm_max	kg/ha	maximum total biomass during timestep
24	bm_grow	kg/ha	total biomass growth during timestep
25	c_gro	kg/ha	total plant carbon growth during timestep

The HRU writer appends `plant_cov` and `mgt_ops` as trailing string fields, just like the water-balance writer.

Aggregation

- Daily values come straight from `hpw_d`.
- Monthly: daily values summed, then divided by days-in-month for the columns that represent averages (`lai`, `tmpav`, etc.), preserving running max for `bm_max` and `lai_max`.
- Yearly: same pattern at year boundary.
- Annual average: divided by `time%yrs_prt` at end of simulation.

Related files

- Water balance outputs.
- Channel and sediment outputs.